Unintended Benefits of Large Irrigation Projects in India

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Abstract

Groundwater is an essential resource, used for both drinking water and agricultural irrigation globally. However, critical groundwater aquifers are being overexploited with particularly severe effects in India. Although India's large-scale irrigation canal projects are designed to drive economic development, they have the potential to also reverse aquifer depletion. We leverage a design feature of the world's largest irrigation canal in a geographic regression discontinuity design to estimate the effects of this canal. We find that the canal surface water causes aquifer levels to rise by 80% of baseline levels, indicating the irrigation canal project effectively reverses the overuse of essential groundwater resources. We further find that surface water access reduces the costs of constructing field-level irrigation infrastructure, saving a total of $13 million over seven years. We also find a 20% increase in irrigated farmland and a 12% increase in planting high-value cash crops.

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1 Introduction

Groundwater is a vital resource for sustaining life and driving agricultural productivity, but it faces an imminent threat from excessive extraction. 36% of global groundwater aquifer systems are being extracted faster than they can be recharged, and overexploitation is accelerating in most of these aquifers (Jasechko et al. 2024). These overdrafted systems include many critical agricultural aquifers, predominately in South Asia and North America (Gleeson et al. 2012). India stands out as a major contributor to this pressing issue, with predictions that environmental flow limits in the country will peak by 2030 (de Graaf et al. 2019). Immediate action is crucial to mitigate severe implications for regional and global food security (Bhattarai et al. 2021; Mukherji 2022) and energy security (Sidhu, Kandlikar, and Ramankutty 2020) and to prevent intensified water-related conflicts, political instability (Gleick, Iceland, and Trivedi 2020; Zaveri et al. 2021; Borgomeo et al. 2021), and increased socioeconomic vulnerability (Fischer et al. 2022). Managing India's heavy dependence on groundwater poses a significant challenge, as the country accounts for a staggering 25% of global groundwater withdrawal, the highest of any single nation (Margat and van der Gun 2013). This intense extraction creates a substantial strain on India's water security, with nearly 60% of irrigation water and 85% of rural drinking water coming from groundwater (CGWB 2022).

India's water management strategy to date has focused on constructing large-scale water infrastructure, such as damming major rivers and routing the water through extensive human-made canals. These projects are intended to boost economic development by providing irrigation and drinking water supplies, ensuring food and energy security, and generating employment opportunities for millions (Shah 2009; Shah et al. 2009); and recent studies find some of these development goals have been met (Duflo and Pande 2007; Blakeslee et al. 2023; Asher et al. 2024). Additionally, although it may not be the intended outcome, such infrastructure projects have the potential to alter groundwater resources in two ways: surface water provides a substitute for groundwater extraction, and the use of the surface water can ultimately recharge the underlying groundwater aquifer. In order to optimally manage these resources, policymakers must understand the interplay between surface water and groundwater systems, crucial for managing India's water-energy-food

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nexus in light of dire predictions of future groundwater depletion (Mukherji 2016, 2022).

In this paper, we examine the impacts of the largest irrigation canal in the world by carrying capacity: the multi-billion-dollar Narmada Main Canal (NMC) in the Indian state of Gujarat. Our research design exploits an important engineering design feature of the NMC. Due to its hydrogeologic characteristics, the NMC is a gravity-fed canal, making surface water available only on one side of the canal. We leverage this unique feature in a geographic regression discontinuity design that effectively compares the units on the canal side with access to surface water to nearly identical units on the other side that lack this resource, providing credible causal inference.

While our primary motivation is to assess the effect of the NMC on groundwater aquifers, we expand our analysis to estimate the impacts of this canal irrigation project on a broad spectrum of outcomes. To do so, we assemble a panel dataset of many outcomes from diverse sources. These data cover the years 2001—when water first started flowing regularly in the NMC—to 2022, with different data sources having different temporal coverage within this time frame. Our groundwater depth data come from India's Central Ground Water Board (CGWB), which reports quarterly in-situ groundwater depth measurements across more than 1400 wells in Gujarat. We also collect village-level data on irrigation infrastructure, irrigated farmland, crop choice, and employment from the decennial Census of India and the septennial Census of Minor Irrigation Schemes. We further add high-resolution remotely sensed data that provide information on annual cropping intensity and proxies for annual economic activity. Using this extensive panel dataset, we analyze not only the impact of the NMC on groundwater depths but also the broader effects of improved water availability on irrigation and agricultural decisions and, ultimately, economic outcomes.

Most importantly, our geographic regression discontinuity reveals that the canal's surface water has caused a significant 80% improvement in average aquifer levels, compared to groundwater depths on the control side of the canal, since water began flowing regularly in 2001. This result highlights the unintended success of the irrigation canal project in mitigating the over-extraction of critical groundwater resources and underscores its potential for broader environmental and economic impacts.

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We then estimate the combined effect of surface water availability and greater groundwater accessibility on agricultural and economic outcomes. We find that villages with access to surface water reduce their expenditures on the construction of groundwater wells and other irrigation infrastructure. The average village saved roughly $33,000 in US dollars over a seven-year period during our sample, or more than $13 million when aggregated over all villages near the canal.[1] In terms of agricultural outcomes, we estimate that in the first decade of water flowing regularly in the NMC, access to this surface water led to a 20% increase in a village's irrigated farmland, coming entirely from converting non-irrigated farmland to canal-irrigated. We further find that these villages are 12% more likely to grow highly profitable cash crops, switching from lower-value drought-tolerant crops. These findings highlight the transformative potential of such large-scale irrigation projects on local agriculture and economies.

This paper makes four contributions to the literature. First, we present the first credible causal estimate showing that large-scale irrigation projects can reverse the depletion of groundwater aquifers, improving the long-run sustainability of this critical resource.[2] To the extent that groundwater sustainability is not the intended result of these projects, this finding provides another high-value benefit for policymakers to consider when designing these large infrastructure projects. Further, our result highlights that optimal water management should jointly consider both surface water and groundwater resources. This finding complements the recent work of Behrer and Pullabhotla (2025) showing that small-scale water infrastructure projects can increase local groundwater recharge rates.

Second, we provide evidence that canal surface water—along with the greater accessibility of groundwater—improves outcomes for the farmers receiving the water. These results include the novel finding that these farmers reduce their expenditure on groundwater irrigation infrastructure by more than $13 million over only seven years—a substantial cost savings for farmers with an

1. These savings occurred from 2007 to 2014. These years are the only period for which the data are available when water was flowing regularly in the NMC.

2. Dinesh Kumar, Jagadeesan, and Sivamohan (2014) present descriptive evidence to suggest that groundwater levels improved after the construction of the NMC. Additionally, prior work has shown how other Indian government policies or programs, such as trade policy (Sekhri 2022) or electricity supply schemes (Badiani, Jessoe, and Plant 2012; Jacoby 2017; Badiani-Magnusson and Jessoe 2019), can impact groundwater extraction and aquifer levels.

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average monthly household income of only $145 in these years[3]—providing yet another unintended benefit of the irrigation projects.[4] We further corroborate prior evidence that water access improves agricultural outcomes in India (Duflo and Pande 2007; Sekhri 2014; Blakeslee et al. 2023; Asher et al. 2024; Boudot-Reddy and Butler 2024).[5] We do not, however, find a meaningful effect on broader economic outcomes, although this may be due to our research design being susceptible to geographic spillovers.[6]

Third, our results highlight that some benefits of large irrigation projects are unevenly distributed. Using our geographic RDD, we identify effects by comparing the side of the NMC with surface water to the side without, so the impacts we estimate are experienced only by farmers on one side and not by their neighbors across the canal. Thus, all of our beneficial results—improvements in groundwater aquifers, reductions in irrigation expenditures, increases in irrigated farmland, and changes in crops—accrue to only a subset of the population represented in our data, raising potential distributional concerns. These results align with Blakeslee et al. (2023) and Asher et al. (2024), who find that these projects have heterogeneous impacts on population and economic development.

Finally, our groundwater findings contribute to an ongoing policy debate about how best to reverse aquifer depletion in Gujarat and throughout India (Kumar and Perry 2019). In addition to irrigation projects like the NMC, Gujarat has also implemented rainwater harvesting schemes (Sekhri 2013) and has altered the supply of electricity for agricultural uses.[7] While some researchers

3. See `https://mospi.gov.in/sites/default/files/publication_reports/nss_rep_576.pdf` for the 2013 report on Income, Expenditure, Productive Assets and Indebtedness of Agricultural Households in India. Average monthly income for agricultural households in 2012–2013 is in Statement 3.5 on page 20. We use the exchange rate of 54.78 Indian rupees per US dollar from the year 2013.

4. Sekhri (2014) and Boudot-Reddy and Butler (2024) show that depth to groundwater impacts groundwater pump adoption, but they do not link this finding to large-scale irrigation projects.

5. Additional work has shown the agricultural benefits of irrigation projects in other contexts, such as sub-Saharan Africa (Jones et al. 2022; Cisse et al. 2025).

6. The existing literature is inconclusive on the extent to which water access alters population and economic outcomes. Sekhri (2014) finds that groundwater access reduces rural poverty rates, but Boudot-Reddy and Butler (2024) do not detect such an effect. Focusing on irrigation canals, Asher et al. (2024) show that these projects induce structural transformation through population changes, while Blakeslee et al. (2023) find that economic and population effects are heterogeneous between towns and villages. In our setting, if the economic benefits of the NMC spill over and accrue to both sides of the canal, then our geographic RDD may not detect differential economic activity on the two sides of the canal.

7. This electricity supply program is known as Jyotigram Yojana, meaning "village lighting scheme." The program's goal is to provide a reliable and consistent supply of three-phase power to rural areas in the state, while also providing a subsidized but limited supply of electricity for agricultural irrigation. This program has provided electricity to

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link this electricity supply scheme to improvements in groundwater resources in Gujarat, others dispute this claim (Shah and Verma 2008; Kumar, Scott, and Singh 2011; Chinnasamy et al. 2021).[8] Our results show unequivocally that large irrigation projects have the potential to substantially improve groundwater aquifers.

2 Background

2.1 Water management in Gujarat

Gujarat has a semi-arid climate requiring irrigation for most agricultural activity. Farmers irrigate using both surface water—from natural bodies of water or human-made canal projects—and groundwater that they pump from underground aquifers. Surface water and groundwater are not only substitutes in agricultural production, but the supply of these two resources is also closely linked. Groundwater undergoes natural recharge from precipitation and artificial sources, including water conservation structures and canal irrigation systems. As a result, an optimal water management strategy requires an integrated approach that considers both surface water and groundwater, as well as the important relationship between them. Further, in India, surface water and groundwater management are inextricably linked to the local political economy, subsistence agriculture, employment prospects, and food security (Shah 2009).

Gujarat has made notable strides in managing groundwater resources and fostering village development (Chindarkar and Grafton 2019). For example, over the last two decades, the number of semi-critical, critical, and over-exploited groundwater sub-districts has declined in Gujarat compared to the national trend (CGWB 2022). The state relies heavily on monsoon rainfall for groundwater recharge, contributing more than 70% of average annual recharge (CGWB 2022). Artificial recharge mechanisms—including water from the Narmada Main Canal—account for the remaining

all villages and improved farm investment (Chindarkar, Chen, and Sathe 2020), and it has indirectly contributed to the provision of village amenities such as primary healthcare units (Chen, Chindarkar, and Xiao 2019). For more program details, see Shah and Verma (2008), Chindarkar and Grafton (2019), and Chinnasamy et al. (2021).

8. Badiani, Jessoe, and Plant (2012), Badiani-Magnusson and Jessoe (2019), and Ryan and Sudarshan (2022) study the effects of subsidizing or limiting the supply of agricultural electricity in other Indian contexts.

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groundwater recharge. However, the full extent of the NMC's effects on the state are not well understood. We seek to estimate how the surface water from the NMC impacts groundwater depth and irrigation practices, as well as broader effects on the agricultural sector and the economy as a whole.

2.2 Narmada Main Canal

India's First Five-Year Plan in 1951 popularized the notion that large dams were the "temples" of democracy, emphasizing that "development through irrigation" was crucial for the primarily agrarian economy (Shah 2009). This development policy stance led to the initiation of large irrigation projects after India achieved independence, with the aim of boosting agricultural production and sustaining employment opportunities in the sector. However, the full potential for irrigation was not initially realized, and the early projects exhibited significant disparities across the country. To address these issues, the government introduced the Command Area Development (CAD) program in 1974. In the 1990s, the program was re-imagined to incorporate more comprehensive water management practices and deliver tangible benefits to farmers throughout the country.

There are currently 150 projects under the CAD program, encompassing a total culturable command area[9] of more than 16 million hectares or roughly 25% of all irrigated farmland in India (Shah 2009). Gujarat has emerged as a leader in these water management projects, leading to strong economic development through industrial and agricultural growth (Shah et al. 2009). In total, the state is home to 39 CAD projects covering a culturable command area of nearly 1.5 million hectares (NWIC 2023). Among these projects in Gujarat, one canal in particular stands out as the largest irrigation canal in the world by carrying capacity: the Narmada Main Canal (NMC).

The NMC begins at the Sardar Sarovar Dam (SSD)—the second largest concrete dam in the world based on total volume—on the Narmada River.[10] This dam was a pivotal element of Gujarat's long-term development policy, with the state government anchoring its crucial policies for agriculture

9. "Culturable command area" describes the area of land that can be irrigated by the program and is suitable for agricultural cultivation.

10. Without this dam, the entirety of the Narmada River would flow into the Arabian Sea without passing through our area of study.

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and industry around this project.[11] As such, this dam was constructed with multiple development purposes in mind. In addition to serving as the reservoir for the NMC, the dam also has 1450 MW of hydroelectric capacity to provide power across four states. The SSD also hosts the Statue of Unity, a cultural landmark that stands as the tallest statue in the world at a height of 182 meters. Construction on the dam began in 1987. From 1999 to 2017, the height of the dam was increased several times, ultimately reaching its current height of 163 meters.

Starting from the SSD, the NMC extends 532 km through Gujarat and into Rajasthan, making it the second longest irrigation canal in India. Figure 1 depicts this path of the canal, beginning near the eastern edge of Gujarat and proceeding north and west to Rajasthan. Water began flowing regularly through the canal in roughly 2001–2002, and the quantity of water supplied by the canal has increased steadily since then (Kumar and Perry 2019). At its maximum, the NMC is designed to carry 9.5 million acre-feet of water annually, providing irrigation to an area of 1.8 million hectares of farmland.[12] Figure 1 also shows the command area that has access to surface water from the canal. The full cost of the canal project is estimated at more than $9 billion US dollars.[13]

An important feature of the NMC is that it is a gravity-fed canal, meaning that its water can be accessed at elevations below the canal but not at higher elevations above it. As a result, the canal water is available to farms on only one side of the canal and not to farms on the other side. Figure 1 depicts this fact, showing that the command area south and west of the canal has access to irrigation water, while the areas of higher elevation to the east and north are outside the command area. Additionally, the exact location of the canal—and the resulting command area—was chosen by engineers and geologists based on the hydrogeologic characteristics of the terrain, generally following a contour of constant elevation. Deviation from this path would have been costly, so the location of the command area was unlikely to be manipulated for local political or economic reasons (Asher

11. Despite the potential benefits of the SSD, its creation was highly controversial. The Narmada Bachao Andolan movement formed to protest the construction of the SSD and other dams on the the Narmada River. Their concerns included the displacement of local populations and the environmental impact of the dams. A full accounting of the benefits and costs of the SSD and the NMC would include the careful consideration of these sensitive topics, but that is beyond the scope of this analysis.

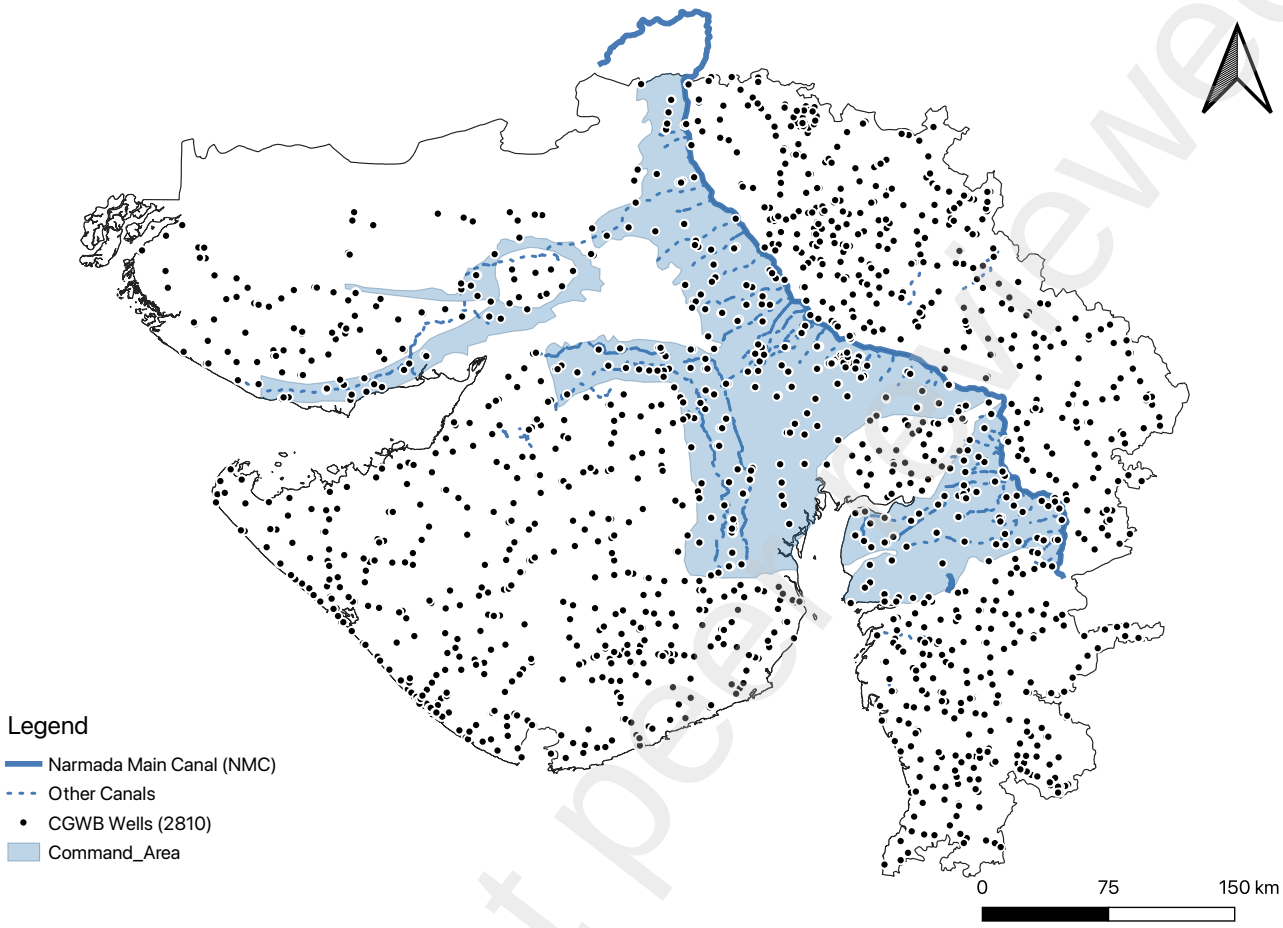
12. See `nca.gov.in/aboutus_nwdt.htm` for these and other details on the Narmada Main Canal.

13. This cost estimate comes from reporting by The Indian Express at `indianexpress.com/article/cities/ahmedabad/narmada-project-will-be-completed-by-2025-at-rs-79000-cr-says-minister-8495015/`.

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Figure 1: Map of Narmada Main Canal and study area



Notes: This map depicts the important features of our study within the state of Gujarat. The solid blue line shows the location of the Narmada Main Canal, beginning at the Sardar Sarovar Dam near the eastern edge of Gujarat, and flowing north and west to Rajasthan. The shaded blue area depicts the NMC's command area, or the area of farmland that has access to surface water from the canal, to the south and west of the NMC. The thinner dashed blue lines show smaller canals that carry water away from the NMC to supply the command area. The black dots show the locations of the Central Ground Water Board's observation wells where groundwater depths are measured.

et al. 2024; Blakeslee et al. 2023). We leverage these facts in our empirical strategy.

3 Empirical strategy

To estimate the causal effects of access to surface water from the NMC project, we exploit the hydrogeologic features of the canal in a geographic regression discontinuity design (RDD). As we describe above and show in Figure 1, surface water is available on only one side of the gravity-fed canal, generating a sharp regression discontinuity at the NMC. This canal, however, would not be

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expected to discontinuously alter other physical characteristics—geographic, geologic, agronomic, or climactic—of the region. Thus, our RDD effectively compares units on the side of the canal with surface water to the nearly identical units on the other side that lack this important resource, isolating the impact of surface water to generate causal estimates.

Our RDD regression specification is

$$Y_{it} = \alpha + \tau T_i + \beta_1 X_i + \beta_2 T_i X_i + \gamma Z_{it} + \varepsilon_{it} \tag{1}$$

where Y_{it} is an outcome—such as groundwater depth—for location i at time t.[14] T_i is a binary variable equal to one if the unit is on the command area side of the canal where surface water is available. X_i is the distance to the NMC;[15] following Gelman and Imbens (2019), we control for a local linear function of the distance to the NMC. Z_{it} is a vector of covariates that we include as controls to improve precision (Calonico et al. 2019).[16] ε_{it} is an idiosyncratic error term.

Our coefficient of interest is τ, which provides the causal effect of access to surface water.[17] Our identifying assumption is that the hydrogeologic characteristics that lead to the placement of the canal are quasi-random, such that no other factors also vary discontinuously at the canal.[18] Our identification is similar to prior research exploiting the placement of canals in a geographic RDD (Jones et al. 2022; Asher et al. 2024) or other hydrogeologic features that quasi-randomly determine access to water (Sekhri 2014; Blakeslee, Fishman, and Srinivasan 2020; Blakeslee et al. 2023; Boudot-Reddy and Butler 2024).

14. Our locations and time periods vary for different outcomes of interest. Locations are either observation wells, villages, or satellite grid cells. Time periods are either quarters or years.

15. We denote the command area side of the canal with a positive distance and the other side of the canal with a negative distance. Then the canal itself, which represents the regression discontinuity cutoff, is equal to a distance of zero.

16. We control for soil and sedimentary deposit thickness, slope, elevation, and temperature. See Section 4.5 for more information on these covariates. Appendix Figure A2 shows that these covariates vary continuously at the canal.

17. Because of our geographic RDD, this estimate is a local average treatment effect relevant at the location of the canal. The relative homogeneity of characteristics—other than the NMC—in this area suggests the effects are applicable more broadly through the region.

18. A potential concern is that the NMC intersects with rivers and discontinuously alters these other sources of surface water. However, the canal is engineered to flow either above or below rivers, never intersecting these other bodies of water. See Appendix A.1 for more information and Appendix Figure A1 for photographic examples.

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To ensure we include comparable units on each side of the NMC, we estimate this regression on the subsample of observations that fall within a specified distance—or bandwidth—from the canal. Due to potential bias and incorrect inference when selecting a bandwidth, we follow the recommendation of Keele and Titiunik (2015) and use the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014). We select our bandwidth using the data-driven optimal bandwidth selection appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020).[19] Note that the optimal bandwidth can vary for different outcomes; for each estimate, we report this bandwidth and the number of treatment and control observations within the bandwidth.

4 Data

In order to estimate the causal effect of surface water from the NMC on a broad set of outcomes, we combine multiple data sources: in-situ measurements of groundwater depths, India's Census of Minor Irrigation Schemes, decennial Census of India, and remotely sensed data products. Here we describe each set of outcomes that we use in our geographic RDD to better understand the full extent of the NMC's impacts.[20]

4.1 Groundwater depth

Our main outcome of interest is the groundwater aquifer level on either side of the NMC. We use groundwater depth data from India's Central Ground Water Board (CGWB), which has conducted nationwide monitoring of representative groundwater observation wells since 1969. The CGWB currently monitors 23,125 such wells through the country with 1,471 wells in Gujarat alone.[21] The CGWB records groundwater depths at these wells four times annually: in January, May, August, and November.

19. We also show some results using a bandwidth of 10 km, which we selected after conducting phone interviews with locals to determine distance of surface water delivery from the NMC.

20. Appendix Table A1 provides a brief summary of these data, including the data source, years covered, and spatial resolution.

21. The number of monitored wells varies modestly over time, and a few wells occasionally have missing observations due to technical limitations and climatic shocks (CGWB 2022).

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We use these quarterly groundwater depth measurements from the observation wells in Gujarat over the years 2001 to 2021, the first 21 years during which water was flowing regularly in the MNC.[22] Figure 2 depicts the average quarterly groundwater depth in Gujarat on each side of the NMC over this time period. Groundwater depths on the control side—where no canal water is available—are consistently deeper than on the treated side with water, and the gaps generally increase over the years. Given the large area of Gujarat, many factors beyond the presence of the canal could explain these observed differences in groundwater depths. In our empirical strategy, we use a geographic RDD to isolate the causal effect of the canal's irrigation water on groundwater depth.

Figure 2 also makes evident that groundwater levels vary over the course of the year: depth to groundwater is typically smallest in quarter 3 and largest in quarter 2. The monsoon season, which facilitates increased groundwater recharge, occurs during quarter 3, so groundwater is closest to the surface during this quarter and then recedes over subsequent quarters. Because the availability of groundwater—and its resulting value to agricultural sector—may vary throughout the year, we estimate causal effects on groundwater depths by quarter of year, as well as the average effect across all quarters.

4.2 Irrigation infrastructure and irrigated farmland

We also examine how the availability of surface water through the NMC—along with any difference in groundwater depth—affects irrigation practices. We look specifically at the irrigation infrastructure that farmers use and the amount of farmland that they irrigate.

We take data on irrigation infrastructure from the Fifth Census of Minor Irrigation Schemes, which was conducted in 2013–2014.[23] This census reports village-level data on the number and cost of different types of field-scale irrigation infrastructure constructed since the prior census in 2006–2007. We focus specifically on the number and cost of groundwater wells constructed over this seven-year time period, when farmers had begun to adapt to the availability of water in the

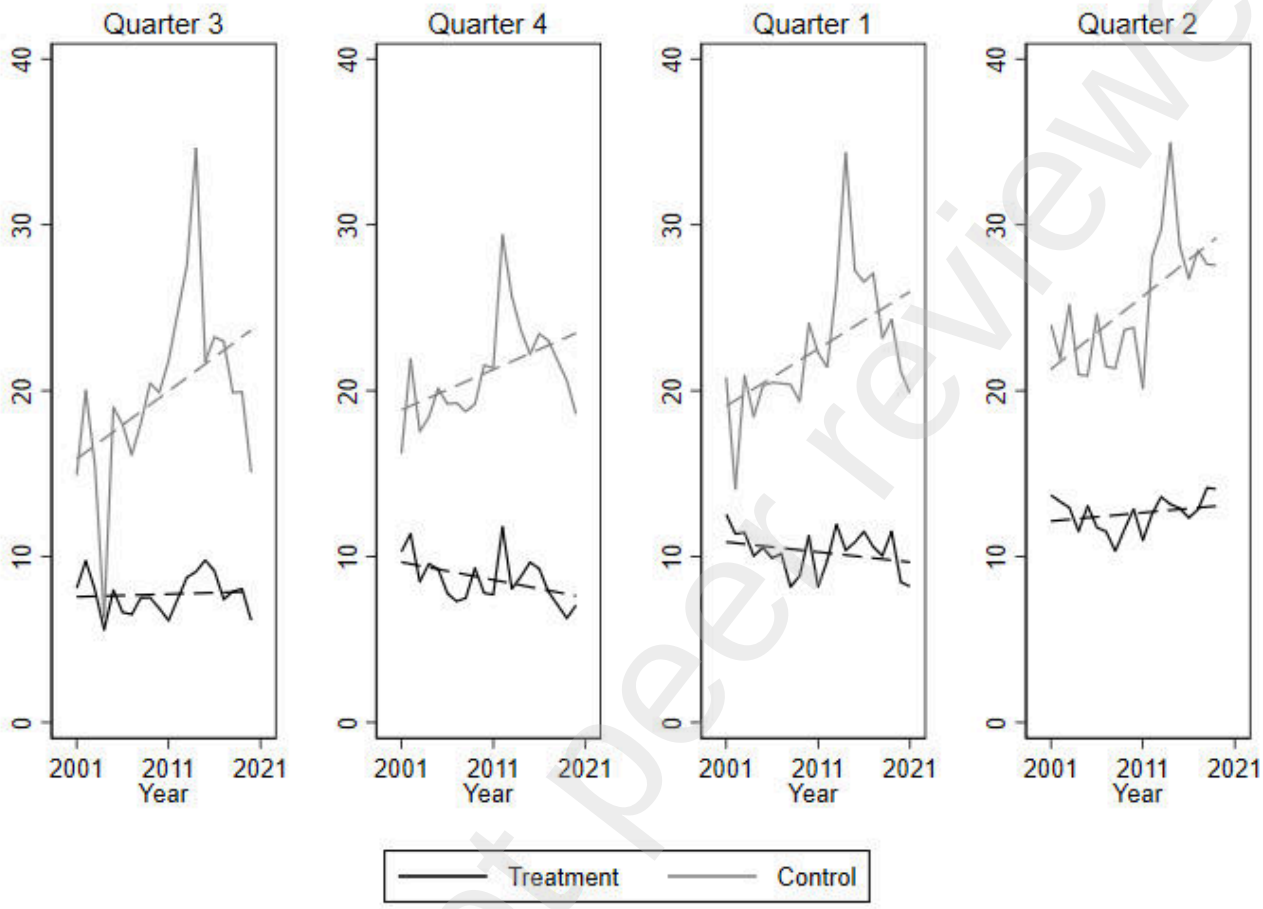
22. The data are available from the India Water Resource Information System (NWIC 2023).

23. The data are available from the India Department of Water Resources, River Development and Ganga Rejuvenation (2014).

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Figure 2: Quarterly groundwater depths on each side of the Narmada Main Canal



Notes: This figure depicts the average quarterly groundwater depth in Gujarat on each side of the NMC over the years 2001 to 2021. Groundwater depth is measured in meters below ground level (mgbl), with a smaller number indicating a shallower aquifer that is closer to the surface. Each panel depicts a different quarter of the year, beginning with quarter 3, which is that start of the monsoon season. Within each panel, the darker line shows groundwater depth for the treatment group on the command area side of the NMC, while the lighter line shows values for the control group on the other side of the NMC. The dashed lines plot fitted linear trends in groundwater depth.

command area.[24]

To examine effects on the amount of irrigated farmland, we use data from the decennial Census of India in 2001 and 2011.[25] This census reports the area of farmland in each village that is irrigated or not irrigated, and it further disaggregates the irrigated land based on the source of irrigation. We use the years 2001 and 2011 to examine how the amount and sources of irrigated farmland changed

24. We further examine effects on the types of groundwater wells: deep tubewells, shallow tubewells, and dug wells. See Appendix Section A.3.2 for these results.

25. We access Census of India data directly from the Office of the Registrar General and Census Commissioner of India (2001, 2011) and through the SHRUG platform (Asher et al. 2021).

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over the first decade of water flowing regularly in the NMC.

4.3 Crop choice and cropping intensity

We further consider how the availability of water—either surface water through the NMC or more accessible groundwater—alter the cropping decisions of farmers. We look at both the kinds of crops that farmers choose to grow and their cropping intensity.

We take crop choice data from the decennial Census of India in 2011.[26] This census reports the top three crops grown in each village, which we use to denote whether a village grows any cash crops, any drought-tolerant crops, or any water-intensive crops.[27] We use these cropping data from 2011—approximately 10 years after water began regularly flowing in the NMC—to estimate the causal effect of surface water availability on these broad cropping patterns.

We use data on cropping intensity from the high-resolution satellite-based dataset of Liu et al. (2021).[28] This dataset reports annual cropping intensity for the years 2001 to 2019 at a gridded resolution of 250 meters. Each year, every grid cell is classified by one of four cropping intensities: none, single, double, or triple cropping. We use all available years of data to examine if the availability of surface water from the NMC causes land to be used more intensively.

4.4 Labor markets and economic activity

We finally estimate the effects of the NMC's surface water on labor markets and the local economy. To measure these outcomes, we consider the number of workers by employment categories, nighttime lights, and ambient human activity.

We use labor market data from the decennial Census of India in 2001 and 2011. This census

26. The 2001 census does not include comparable data on crop choice.

27. We follow Boudot-Reddy and Butler (2024) in classifying crops. We denote a village as growing cash crops if it grows sugarcane, oilseed, cotton, or tobacco. We denote a village as growing drought-tolerant crops if it grows millet, sorghum, maize, pigeon pea, or groundnut. We denote a village as growing water-intensive crops if it grows sugarcane, cotton, or rice. Note that these cropping categories are neither exhaustive nor mutually exclusive, so a village could be characterized as growing zero, one, two, or all three crop types.

28. Liu et al. (2021) use the enhanced vegetation index (EVI) from the Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices (MOD13Q1) to classify land based on the number of annual cropping peaks: zero, one, two, or three.

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reports the total number of workers in each village, as well as the number of workers disaggregated into different employment categories. We focus on two categories in particular: landless "agricultural workers" employed by others and land-owning "cultivators" who farm their own land. We use these data for 2001 and 2011 to examine how labor—and agricultural labor specifically—changed over the first decade of regular NMC water availability.

As a proxy for economic activity, we use the nighttime lights dataset of Li et al. (2020).[29] This dataset reports the annual average brightness of lights at nighttime at a gridded resolution of approximately 1 km.[30] Nighttime lights have been used as a high-resolution proxy for economic activity and growth in settings without such administrative data (Henderson, Storeygard, and Weil 2012). We use these data for the years 2001–2018 to examine if the availability of canal water has caused differential economic growth.

We look for human activity using ambient population data from the LandScan Global Population Database.[31] This high-resolution data product reports an annual average measure of ambient population at a gridded resolution of approximately 1 km.[32] These ambient population data are intended to capture the full number of people who pass through a given space, rather than recording only residents. We use these data for the years 2001–2022 to estimate the average effect on human activity since water began flowing regularly in the NMC.

4.5 Control variables

In order to improve the precision of our estimates, we control for hydrogeologic, topographic, and climactic characteristics that could impact groundwater depth and other outcomes of interest (Calonico et al. 2019). We include five such covariates: soil and sedimentary deposit thickness,

29. Li et al. (2020) harmonize existing nighttime light data from the Defense Meteorological Satellite Program and the Visible Infrared Imaging Radiometer Suite to construct a comprehensive dataset of nighttime lights for the years 1992–2018.

30. Li et al. (2020) report data using a grid of 30 arc-seconds. At the equator, 30 arc-seconds are equivalent to nearly 930 meters, or approximately 1 km.

31. These data are available from Sims et al. (2022). LandScan combines high-resolution, remotely-sensed geospatial data with machine learning algorithms to project ambient population.

32. As with the nighttime lights data, LandScan report ambient population using a grid of 30 arc-seconds, which is approximately 1 km at the equator.

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two measures of slope, elevation, and the average annual of monthly maximum temperature.[33]

The availability and recharge of groundwater resources at a particular location depend on hydrogeologic and topographic features of the landscape (CGWB 2022). We draw from two data sources to control for these features. First, Pelletier et al. (2016) report on permeable layers above bedrock at a gridded resolution of approximately 1 km. Using this dataset, we control for the thickness of soil and sedimentary deposit and the fraction of land that is classified as a "hillslope."[34] Second, EROS (2022) reports land elevation at a gridded resolution of approximately 30 m, which we include as a covariate. We further use these elevation data to calculate the average slope in each grid cell, which we also include as a control.[35]

Climate and weather can also affect the recharge of groundwater, as well as agricultural uses of the land. We take weather data from TerraClimate (Abatzoglou et al. 2018), which reports monthly weather outcomes at a gridded resolution of 4 km. Because hot temperatures are particularly harmful to agriculture (Hultgren et al. 2025), we use the maximum temperature for every month and average up to the annual level. We include this annual average of monthly maximum temperature as our final covariate.

5 Results and Discussion

5.1 Groundwater depth

Our primary outcome of interest is the depth to groundwater, measured as meters below ground level, with smaller values indicating that groundwater is closer to the surface. Figure 3 displays

33. Appendix Figure A2 shows that these covariates do not exhibit meaningful discontinuities at the NMC. Each of these covariates comes from a gridded data product that does not spatially mirror our outcome variables, so we must construct the corresponding control variable for each outcome variable. In the groundwater depth regressions, we match each observation well to its covariate grid cell and use that grid cell's value. For the other outcomes—which are at the village- or grid cell-level—we take an area-weighted average of all covariate grid cells that overlap with the outcome's spatial area.

34. Land is classified as either "hillslope" or "valley bottom." Pelletier et al. (2016) state that the vast majority of most landscapes is hillslope, so this control often takes a value very close to one.

35. Slope is measured in degrees, ranging from 0 to 90. A slope of 0 degrees represents a flat surface, while a slope of 90 degrees indicates a sheer vertical cliff.

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regression discontinuity (RD) plots for groundwater depth. Panel (a) uses the annual average depth at each observation well, while the other panels use depth measurements for specific quarters of the year. All five plots show evidence that groundwater depths change discontinuously at the canal and are closer to the surface on the treated side of the canal.[36]

Table 1 reports our RD estimates for groundwater depth. The availability of surface water from the NMC causes annual average groundwater depth to be 23.55 meters shallower on the treated side of the canal, as shown in Column (1). The mean depth to groundwater within the RD bandwidth on the control side is 28.74 meters, meaning the NMC causes groundwater depth to improve by more than 80%. This shallower groundwater level directly indicates that surface water from the NMC improves the long-run supply and sustainability of these important groundwater resources. It additionally suggests that farmers can access this resource more easily and at lower cost, which may improve outcomes in this sector and which we will examine further.

Table 1: Regression discontinuity estimates for groundwater depth

	(1) Yearly	(2) Quarter 3	(3) Quarter 4	(4) Quarter 1	(5) Quarter 2
Treatment	-23.55^{***}	-15.54^{***}	-18.69^{***}	-16.82^{***}	-17.25^{***}
	(5.53)	(5.88)	(3.96)	(5.64)	(5.04)
Control mean	28.74	25.00	27.06	28.18	28.98
Bandwidth (km)	9.96	12.57	10.20	14.41	12.07
Control obs.	675	606	551	799	612
Treatment obs.	418	404	339	594	417

Notes: This table reports results of estimating Equation (1) with groundwater depth as the outcome variable. Groundwater depth is measured in meters below ground level (mbgl) with smaller numbers indicating groundwater that is closer to the surface. Column (1) uses the annual average groundwater depth at each observation well. Columns (2)–(5) use groundwater depths for a particular quarter of the year, as given in the column headers. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples and are robust to errors clustered at the observation well. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Columns (2)–(5) of Table 1 report effects in individual quarters of the year, beginning with the monsoon season of quarter 3 and progressing to the driest season in quarter 2. Our point estimate

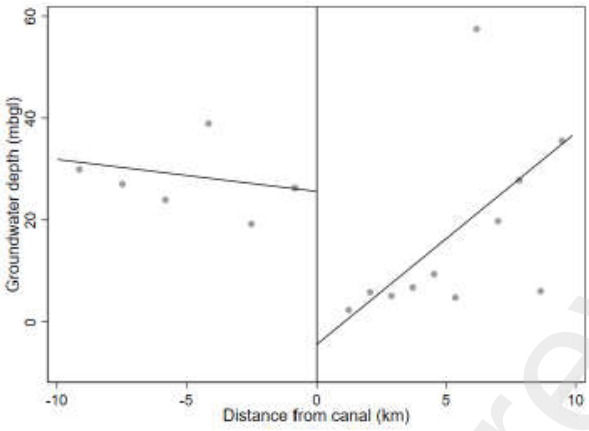
36. We denote the treated side of the canal with positive distance values.

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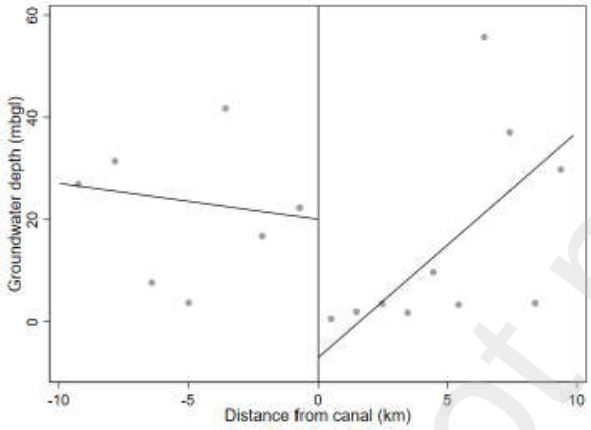
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Figure 3: Regression discontinuity plots for groundwater depth

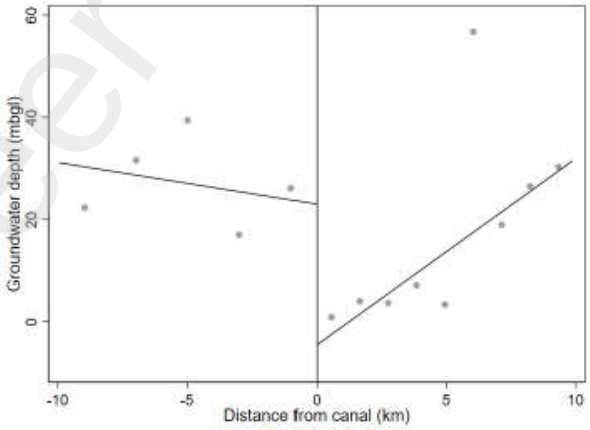
(a) Yearly



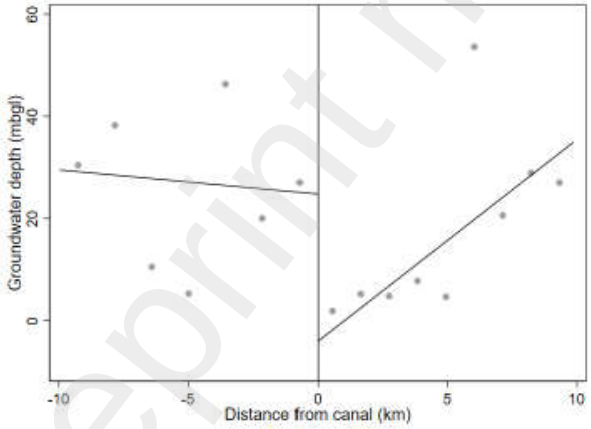
(b) Quarter 3



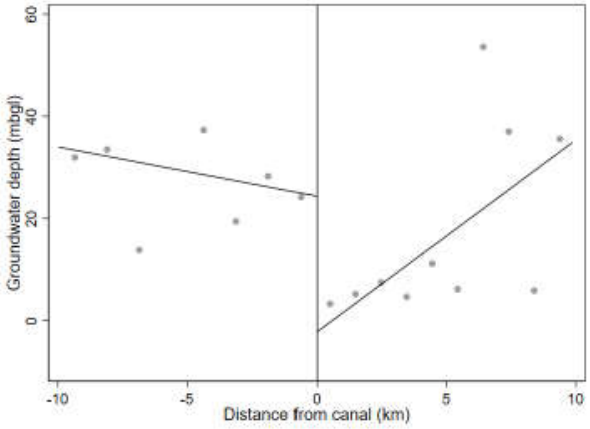
(c) Quarter 4



(d) Quarter 1



(e) Quarter 2



Notes: This figure shows regression discontinuity plots for groundwater depth. Panel (a) includes groundwater depth observations from all four quarters, and the remaining panels show depth by quarter of year. Quarter 3—in Panel (b)—includes the monsoon season, when groundwater depth is typically at its lowest; groundwater levels generally decline over subsequent quarters until the next year's monsoon season. In each figure, we use observations within 10 km of the NMC, approximately equal to the bandwidth of our main yearly groundwater RD from Column (1) of Table 1. Each point represents a bin of observations and depicts the mean of those binned observations. We use the data-driven bin selection procedure of Calonico, Cattaneo, and Titiunik (2015) for optimal evenly spaced bins using spacing estimators. The lines depict the local linear fits on each side of the NMC.

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is smallest in magnitude in quarter 3 when water is typically most plentiful, and the estimates are larger in the latter quarters as water becomes more scarce. Thus, we find that the NMC generally improves the sustainability and accessibility of groundwater by greater amounts when water is particularly scarce and valuable.[37]

One important consideration when interpreting these groundwater results is that groundwater on each side of the NMC is connected through underground aquifers. Over a long enough time horizon, groundwater levels will re-equilibrate with groundwater flowing under the canal from the treated side to the control side. Any re-equilibration that may have occurred during our time period would reduce the groundwater depth differential between the two sides of the NMC, biasing our estimates toward zero. Thus, in the presence of aquifer re-equilibration, our results provide conservative estimates of the causal effect of surface water from the NMC.[38]

5.2 Irrigation infrastructure and irrigated farmland

We next examine how the improved water access due to the NMC—both from the canal water itself and from more accessible groundwater—affects the irrigation decisions of farmers. Table 2 reports our RD estimates on the construction of irrigation wells at the village level during the years 2007–2014, when farmers would have begun adapting to the newly available water resources. Column (1) reports a linear probability model of whether any irrigation wells were constructed in the village during this time, and we find that a village was 11 percentage points less likely to construct a well if it has water from the NMC. Further, those treated villages construct fewer wells on average, although this estimate in Column (2) is imprecise. Finally, as shown in Column (3), the average treated village spent 1.37 million Indian rupees—or roughly $33,000 in US dollars—less on constructing irrigation wells during these years, which is nearly a 70% reduction from the control

37. Appendix Table A2 reports these quarterly results using a consistent bandwidth of 10 km, rather than data-driven optimal bandwidths that vary by quarter. With that consistent bandwidth, this trend becomes even more apparent.

38. As a robustness check, we also estimate effects using only the observation wells within the major aquifer in this region. With this sample restriction, we identify effects using only within-aquifer variation, which would be particularly susceptible to this aquifer re-equilibration concern. Even with this sample restriction, the effect on annual average groundwater depth is of comparable magnitude. See Appendix Table A3 for these results.

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group mean;[39] when we aggregate over all treated villages near the canal, the NMC reduced well construction costs by more than $13 million US dollars over only these seven years.[40]

Table 2: Regression discontinuity estimates for irrigation infrastructure

	(1) Binary	(2) Number	(3) Cost
Treatment	-0.11^{*}	-0.86	-1367.1^{***}
	(0.063)	(1.06)	(375.2)
Control mean	0.70	7.13	1991.4
Bandwidth (km)	25.82	32.52	40.01
Control obs.	1385	1727	2155
Treatment obs.	974	1183	1414

Notes: This table reports results of estimating Equation (1) with village-level measures of irrigation infrastructure construction as the outcome variable. Column (1) uses a binary indicator if a village constructed any irrigation wells during the years 2007–2014. Column (2) uses the number of irrigation wells constructed over these years. Column (3) uses the total cost of irrigation well construction, measured in thousands of Indian rupees. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3 reports our RD estimates for irrigated farmland in the years 2001 and 2011. Looking first at the year 2001, or roughly when water was beginning to flow regularly in the NMC, we find no statistically distinguishable differences in irrigation practices on either side of the canal; these results are in Columns (1), (3), (5), and (7). After a decade of water regularly flowing in the NMC, we find that the area of irrigated farmland was 6.8 percentage points greater in treated villages in 2011, representing roughly a 20% increase, as shown in Column (2). The entirety of this effect comes from greater canal irrigation, as we find no difference in well irrigation, shown in Columns (4) and (6), respectively. Finally, we find less non-irrigated farmland in treated villages, suggesting that the greater irrigation comes from the conversion of non-irrigated farmland, rather than bringing new

39. Appendix Tables A4 and A5 report comparable results by type of irrigation well. Although estimates are imprecise at this more granular level, they suggest that the largest cost savings come from constructing fewer deep tubewells—the deepest and most costly groundwater wells—which is consistent with having shallower and more accessible groundwater.

40. We use the exchange rate of 41.35 Indian rupees per US dollar from the year 2007. To construct the aggregate value, we multiply the per-village value by the 401 villages on the treated side of the canal within 9.96 km, which is the optimal bandwidth for the groundwater depth from Column (1) of Table 1. If irrigation infrastructure cost savings persist at greater distances from the canal, these aggregate savings would be larger.

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land into agriculture.

Table 3: Regression discontinuity estimates for irrigated farmland

	Irrigated		Canal-irrigated		Well-irrigated		Non-irrigated	
	(1) 2001	(2) 2011	(3) 2001	(4) 2011	(5) 2001	(6) 2011	(7) 2001	(8) 2011
Treatment	1.98	6.80**	−0.69	7.79***	2.03	0.48	−2.30	−9.85***
	(3.33)	(3.00)	(0.84)	(1.90)	(3.23)	(3.17)	(3.64)	(3.39)
Control mean	37.73	35.02	3.99	4.65	26.99	29.87	42.90	37.89
Bandwidth (km)	11.78	18.79	32.99	27.03	11.80	14.98	12.64	16.45
Control obs.	805	1288	2187	1804	809	1034	874	1132
Treatment obs.	764	1164	1887	1603	766	945	810	1027

Notes: This table reports results of estimating Equation (1) with village-level measures of farmland as the outcome variable. Each outcome is measured as the percent of total village area. Columns (1) and (2) use area of irrigated farmland, Columns (3) and (4) use area of farmland irrigated by canals, Columns (5) and (6) use area of farmland irrigated by wells, and Columns (7) and (8) use area of farmland not irrigated. Columns (1), (3), (5), and (7) use data from 2001, roughly when the NMC began flowing. Columns (2), (4), (6), and (8) use data from a decade later in 2011. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In summary, we find that the availability of surface water from the NMC, along with shallower and more accessible groundwater, changes irrigation practices in ways that are beneficial to farmers. They saved more than $13 million US dollars over only seven years by constructing fewer and less costly irrigation infrastructure. They further converted some non-irrigated farmland to irrigated, which would be expected to improve crop yields and revenues. Interestingly, the amount of well-irrigated farmland was not affected by the NMC; canal irrigation does not crowd out well irrigation nor does more accessible groundwater increase well-irrigated farmland.

5.3 Crop choice and cropping intensity

We next estimate how these differences in water availability and irrigation practices affect the cropping choices of farmers. Table 4 reports our RD estimates for crop choice in the year 2011. Each column represents a linear probability model of whether a village's top three crops include crops in a particular category. We estimate that a village is 12.3 percentage points more likely

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to grow a cash crop if it receives water from the NMC, which is more than a 40% increase in cash cropping compared to the mean of the control side of the canal, as shown in Column (1). We find a comparable reduction in drought-tolerant crops—treated villages are 12.9 percentage points less likely to grow these crops—as shown in Column (2). Surprisingly, we find no difference in water-intensive crops, as shown in Column (3). Thus, we find that the NMC enables farmers to increase investment in the production of valuable cash crops while reducing their reliance on drought-tolerance crops that tend to be less profitable.

Table 4: Regression discontinuity estimates for crop choice

	(1) Cash crop	(2) Drought-tolerant	(3) Water-intensive
Treatment	0.123^{***}	-0.129^{***}	0.015
	(0.046)	(0.044)	(0.045)
Control mean	0.30	0.42	0.46
Bandwidth (km)	31.70	38.71	34.06
Control obs.	2024	2455	2175
Treatment obs.	1771	2075	1867

Notes: This table reports results of estimating Equation (1) with village-level crop choice in 2011 as the outcome variable. Column (1) uses a binary indicator if a village's top three crops includes a cash crop. Column (2) uses a binary indicator if a village's top three crops includes a drought-tolerant crop. Column (3) uses a binary indicator if a village's top three crops includes a water-intensive crop. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5 reports our RD estimates for annual cropping intensity. Each column gives the results of a linear probability model of whether a satellite grid cell exhibits no cropping, single cropping, double cropping, or triple cropping throughout the year. We find small differences of less than 1 percentage point for each cropping intensity category. For the first three categories, these small effect sizes represent differences of roughly 2% or less between the treated and control sides of the NMC, indicating little effect of improved water access. For triple cropping, though, only 4% of land on the control side is cropped at such a high intensity, so a 0.656-percentage point difference indicates that triple cropping is 16% more likely due to the NMC. Hence, not only do farmers

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with improved water access grow more valuable crops, but some also use that water to increase the number of plantings and harvests throughout a year.

Table 5: Regression discontinuity estimates for cropping intensity

	(1) None	(2) Single	(3) Double	(4) Triple
Treatment	-0.00150^{***}	0.00474^{***}	-0.00766^{***}	0.00656^{***}
	(0.00002)	(0.00003)	(0.00003)	(0.00001)
Control mean	0.12	0.50	0.33	0.04
Bandwidth (km)	2.51	2.59	2.88	3.41
Control obs.	399,000	412,262	454,879	531,848
Treatment obs.	340,347	352,336	392,749	468,103

Notes: This table reports results of estimating Equation (1) with satellite-based cropping intensity in 2001–2019 as the outcome variable. Each column uses a binary indicator if a satellite grid cell's annual observation shows no cropping, single cropping, double cropping, or triple cropping, as given in the column headers. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples and are robust to errors clustered at the satellite grid cell. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.4 Labor markets and economic activity

We finally examine how the availability of water from the NMC—along with all of its additional effects that we have presented—impacts labor markets and economic activity. Table 6 reports our RD estimates for number of laborers in three labor categories for the years 2001 and 2011. We find that in 2001, roughly when water was beginning to flow regularly in the NMC, there were no statistically distinguishable differences in the total number of workers, the number of agricultural workers, or the number of cultivators, as shown in Columns (1), (3), and (5), respectively. Looking at 2011, after a decade of water flowing in the NMC, we estimate that water from the NMC generated an additional 1.9 agricultural workers per village, or a 20% increase compared to the mean of control villages, as shown in Column (4). We find no effect on total workers, as shown in Column (2), indicating that employment must be reduced in non-agricultural sectors. Further, we find no effect on cultivators, as shown in Column (6), indicating that the NMC does not induce

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more landowners to cultivate their land.

Table 6: Regression discontinuity estimates for labor markets

	Total workers		Agricultural workers		Cultivators	
	(1)	(2)	(3)	(4)	(5)	(6)
	2001	2011	2001	2011	2001	2011
Treatment	−0.44	0.49	0.54	1.90*	−0.11	−0.38
	(0.83)	(0.98)	(0.81)	(1.00)	(0.77)	(0.83)
Control mean	52.41	47.91	8.04	10.91	17.97	15.92
Bandwidth (km)	34.01	40.24	38.83	30.97	30.31	30.17
Control obs.	2294	2693	2608	2095	2050	2046
Treatment obs.	1937	2217	2157	1796	1766	1761

Notes: This table reports results of estimating Equation (1) with village-level measures of laborers as the outcome variable. Each outcome is measured as the percent of total village population. Columns (1) and (2) use total number of workers, Columns (3) and (4) use number of agricultural workers, and Columns (5) and (6) use number of cultivators. Columns (1), (3), and (5) use data from 2001, roughly when the NMC began flowing. Columns (2), (4), and (6) use data from a decade later in 2011. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7 reports our RD estimates for nighttime lights and ambient population, which we use as proxies for economic activity and human activity, respectively. For nighttime lights, in column (1), while we can distinguish a statistical difference between the treated and control sides of the canal, this difference is less than 0.01% of the control mean and not economically meaningful. Surprisingly, for ambient population, in column (2), we find that the treated side of the NMC experiences less human activity than the control side. Taking these results together, the treated side of the NMC maintains a comparable level of economic activity among a lower ambient population, as compared to the control side, potentially indicating a greater concentration of income or wealth due to the surface water of the NMC.

When interpreting these results, it is important to remember that workers, human activity, and economic activity can all easily cross the canal. For example, farms on the treated side of the NMC may employ agricultural workers from the control side, which would increase most of these metrics—all but the number of cultivators—on the control side of the NMC. Because of these geographic

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Table 7: Regression discontinuity estimates for economic activity

	(1) Nighttime lights	(2) Ambient population
Treatment	0.025^{***}	-90.8^{***}
	(0.000)	(0.029)
Control mean	10.28	444.4
Bandwidth (km)	3.20	5.98
Control obs.	25,920	69,190
Treatment obs.	26,010	71,704

Notes: This table reports results of estimating Equation (1) with satellite-based measures of economic or human activity as the outcome variable. Column (1) uses annual nighttime lights data for the years 2001–2018. Column (2) uses annual ambient population data for the years 2001–2022. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples and are robust to errors clustered at the satellite grid cell. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

spillovers, our estimates provide lower bounds on the causal effects of surface water from the NMC. It may be the case that the NMC does improve aggregate economic outcomes in a meaningful way, but those positive effects occur on both sides of the canal, so we do not detect them using our geographic RDD.

5.5 Policy discussion

These results have several important policy implications for the design of large-scale irrigation projects. First, we show that surface water from such a project causes a substantial recovery in groundwater depth, improving the long-run sustainability of this valuable resource. This finding is particularly relevant in Gujarat, where an ongoing debate has questioned the efficacy of irrigation projects—compared to other policies, such as electricity supply schemes—in improving the state's groundwater aquifers (Kumar and Perry 2019). Our finding not only shows irrigation projects to be an effective policy tool, but it also highlights that optimal water policy decisions should jointly consider both surface water and groundwater resources.

Second, we find that these irrigation projects can have potentially unexpected impacts on farm-

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ers' irrigation choices and expenditures, providing additional benefits attributable to the project. In our specific setting, we conservatively estimate that farmers reduced irrigation infrastructure expenditures by more than $13 million US dollars over the course of only seven years; if we extrapolate this result to the 24 years that water has flowed regularly in the NMC, farmers near the canal have saved more than $45 million. Given that these farmers had an average monthly household income of $145 within those seven years and rely on this small-scale subsistence agriculture for their livelihood, these reduced expenditures represent a substantial financial benefit accruing to a marginalized and at-risk population.

Third, we find that the irrigation project allows farmers to make more profitable cropping decisions (Blakeslee et al. 2023; Asher et al. 2024). Because of the supply of canal water and the greater availability of groundwater, farmers can access water more consistently and at lower cost. As a result, they switch from drought-tolerant crops to more valuable cash crops—namely, sugarcane, oilseed, cotton, and tobacco in this setting. While we are unable to put a specific value on the increased profits, it again represents a benefit accruing to this distressed population of primarily subsistence farmers, which should be considered when designing such irrigation projects.

6 Conclusion

We use a geographic regression discontinuity design to estimate the impacts—some potentially unexpected by policymakers—of the largest irrigation canal in the world by carrying capacity: the Narmada Main Canal in the Indian state of Gujarat. Most importantly, we find that the regular supply of surface water improves groundwater levels by 80%. This increased availability and accessibility of both surface water and groundwater allows farmers to irrigate more land at lower cost, saving more than $13 million US dollars over only seven years. Some farmers also change their crops, switching from lower-value drought-tolerant crops to more profitable cash crops. However, we estimate only a small impact on agricultural labor markets, and we do not find a meaningful effect on overall economic activity, but both of these outcomes are potentially subject

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to geographic spillovers that would bias our results toward zero.

Large irrigation projects, including the NMC, are designed to be drivers of economic development by providing a regular supply of agricultural water for small-scale and subsistence farmers. This work shows that there are additional benefits beyond simply improving crop yields, including the recovery of groundwater aquifer resources, less expenditure on groundwater irrigation infrastructure, and changes in cropping. These results emphasize that optimal water policy should jointly manage both surface water and groundwater resources and that policymakers should consider these additional benefits when designing future irrigation projects.

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Unintended Benefits of Large Irrigation Projects in India

Supplementary Appendix: For Online Publication

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A.1 Narmada Main Canal

A potential threat to our causal identification is that the Narmada Main Canal must cross rivers and other bodies of water. If the waters of the NMC were to intersect with these others waters, it could discontinuously alter the flow of these rivers. Then our estimated effects would include both the direct effect of the NMC and the additional effect of the NMC altering these others waters.

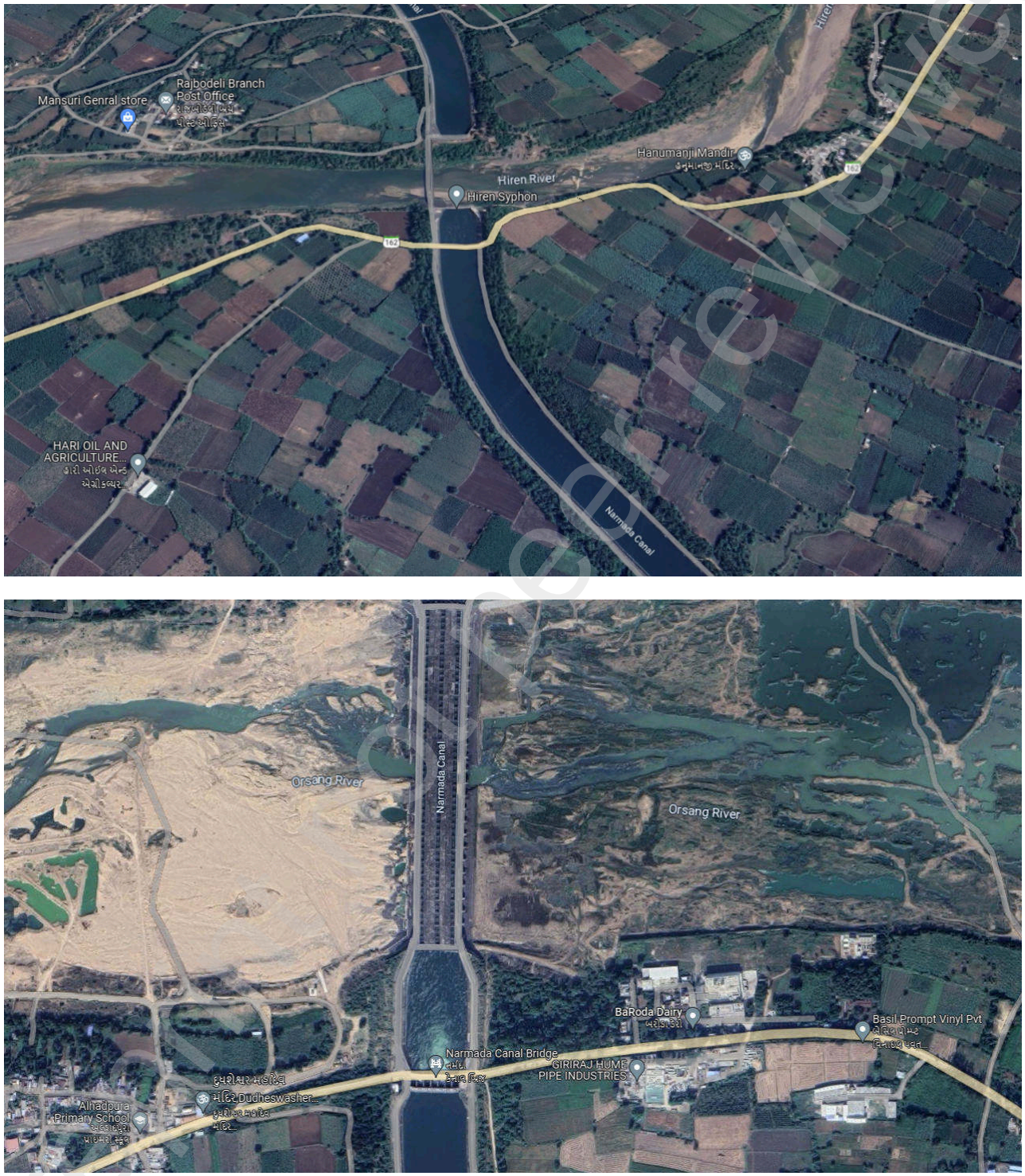
This is not a concern in our case, however, because the NMC is designed to never directly interact with these other bodies of water. When the NMC crosses a river, it is engineered to flow either above or below the river without the waters touching. Figure A1 provides examples, showing the NMC flowing through a canal tunnel or a covered aqueduct to ensure the water does not impact the rivers it is crossing.

A.2 Data

For this analysis, we combine multiple data sources: in-situ measurements of groundwater depths, India's Census of Minor Irrigation Schemes, decennial Census of India, and remotely sensed data products. Table A1 lists the data we use as outcomes or control variables and provides the data source, years covered, and spatial resolution of the data.

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Figure A1: Narmada Main Canal crossing rivers without intersecting



Notes: Both images are from Google Maps and show examples of the NMC crossing rivers without their waters intersecting. The top image shows the NMC crossing the Hiren River; the NMC is engineered to flow through a tunnel underneath the Hiren River without their waters interacting. The bottom image shows the NMC crossing the Orsang River; the NMC is engineered to flow through a covered aqueduct above the Orsang River without their waters interacting.

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Table A1: Data summary

Data	Source	Years	Spatial Resolution
Outcome Variables			
Groundwater depth	India WRIS[a]	2001–2021	Well
Irrigation infrastructure	Fifth MI Census[b]	2007–2014	Village
Irrigated farmland	Census of India[c]	2001 & 2011	Village
Crop choice	Census of India[c]	2011	Village
Cropping intensity	Terra MODIS[d]	2001–2019	250 m
Labor markets	Census of India[c]	2001 & 2011	Village
Nighttime lights	NOAA DMSP & VIIRS[e]	2001–2018	1 km
Ambient population	LandScan[f]	2001–2022	1 km
Resident population	Census of India[c]	2001 & 2011	Village
Control Variables			
SSD thickness	NASA ESDIS[g]	NA	1 km
Hillslope	NASA ESDIS[g]	NA	1 km
Elevation	USGS SRTM[h]	NA	30 m
Slope	USGS SRTM[h]	NA	30 m
Maximum temperature	TerraClimate[i]	2001–2022	4 km

[a] NWIC (2023)
[b] Department of Water Resources, River Development and Ganga Rejuvenation (2014)
[c] Office of the Registrar General and Census Commissioner of India (2001, 2011)
[d] Liu et al. (2021)
[e] Li et al. (2020)
[f] Sims et al. (2022)
[g] Pelletier et al. (2016)
[h] EROS (2022)
[i] Abatzoglou et al. (2018)

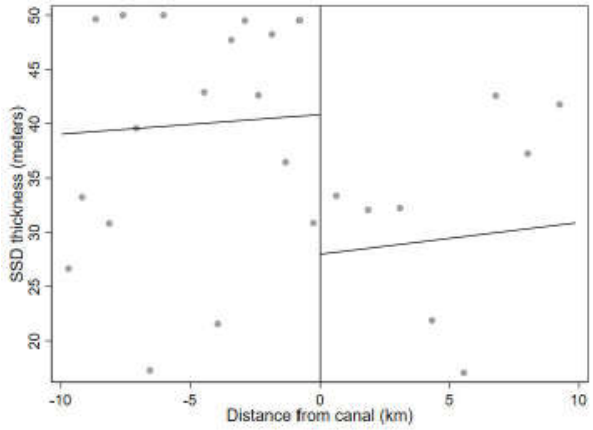
As we describe in Section 4.5, we include five control variables to improve the precision of our regression discontinuity estimates. These variables are soil and sedimentary deposit thickness, two measures of slope, elevation, and the annual average of monthly maximum temperature. Figure A2 depicts regression discontinuity plots for each of these covariates. As expected, we find that none of these covariates exhibit a substantial discontinuity at the canal.

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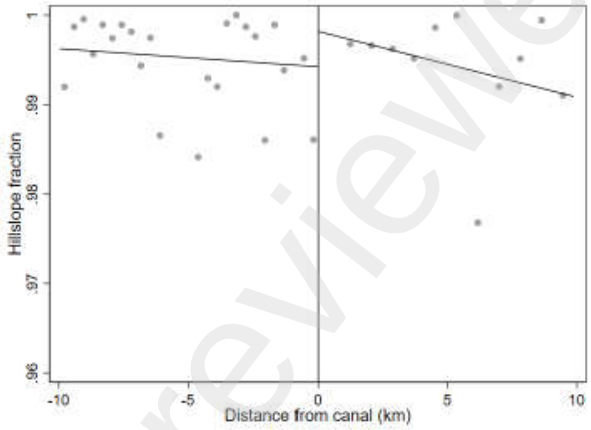
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Figure A2: Regression discontinuity plots for covariates

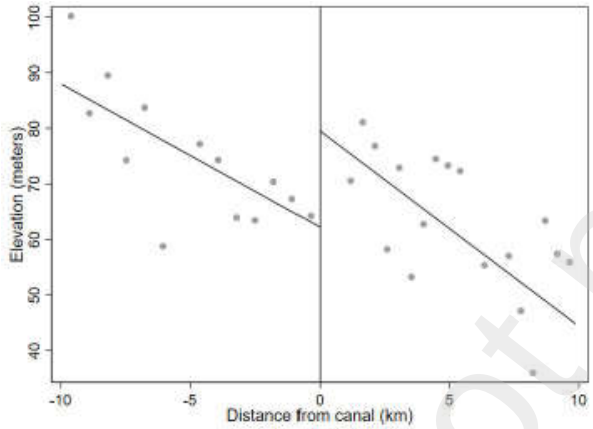
(a) Soil and sediment deposit thickness



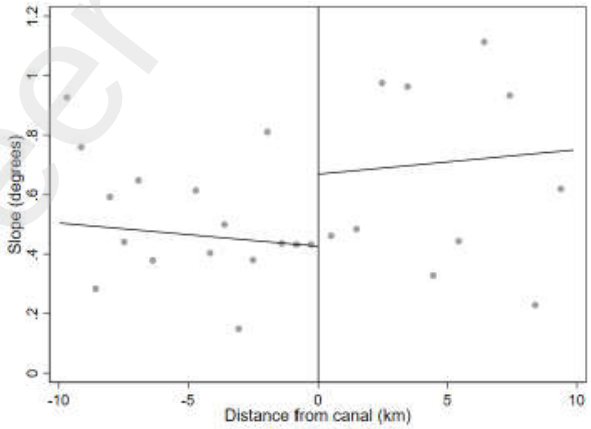
(b) Hillslope fraction



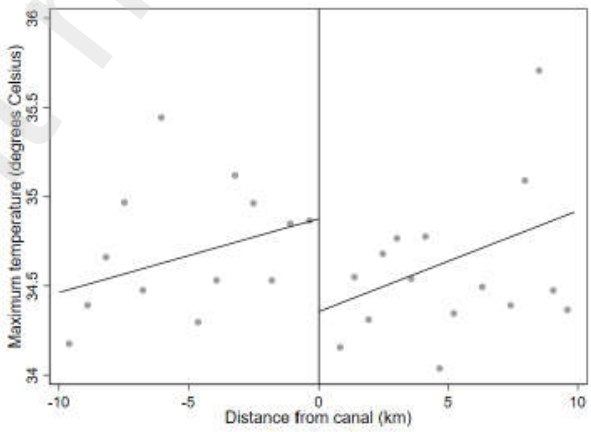
(c) Elevation



(d) Slope



(e) Maximum temperature



Notes: This figure shows regression discontinuity plots for each of the five covariates that we include as control variables to improve precision. In each figure, we use observations within 10 km of the NMC, approximately equal to the bandwidth of our main yearly groundwater RD from Column (1) of Table 1. Each point represents a bin of observations and depicts the mean of those binned observations. We use the data-driven bin selection procedure of Calonico, Cattaneo, and Titiunik (2015) for optimal evenly spaced bins using spacing estimators. The lines depict the local linear fits on each side of the NMC.

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A.3 Results

A.3.1 Robustness of groundwater results

In our main RD estimates of Equation (1), we select a bandwidth around the NMC using the data-driven optimal bandwidth selection procedure of Calonico, Cattaneo, and Farrell (2020). As we show in Table 1, this procedure results in different bandwidths for different groundwater measurements: 8.64 km for annual average groundwater measurements and bandwidths of 9.03–24.27 km for measurements in individual quarters. To show the robustness of these results to alternate bandwidths, we instead set a bandwidth of 10 km, based on phone interviews with locals to determine distance of surface water delivery from the NMC. Table A2 reports these results using a bandwidth of 10 km. We find that effects are comparable in magnitude to the main results in Table 1.

Table A2: Regression discontinuity estimates for groundwater depth with alternate bandwidth

	(1) Yearly	(2) Quarter 3	(3) Quarter 4	(4) Quarter 1	(5) Quarter 2
Treatment	-23.42^{***}	-16.86^{***}	-18.85^{***}	-20.06^{***}	-18.23^{***}
	(5.51)	(4.55)	(3.93)	(5.22)	(3.92)
Control Mean	28.05	23.03	25.89	27.05	27.80
Bandwidth (km)	10	10	10	10	10
Control obs.	696	506	538	539	525
Treatment obs.	418	285	317	331	310

*Notes:*This table reports results of estimating Equation (1) with groundwater depth as the outcome variable. Groundwater depth is measured in meters below ground level (mbgl) with smaller numbers indicating groundwater that is closer to the surface. Column (1) uses the annual average groundwater depth at each observation well. Columns (2)–(5) use groundwater depths for a particular quarter of the year, as given in the column headers. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples and are robust to errors clustered at the observation well. We use a bandwidth of 10 km in all regressions, and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

One potential concern with our RD estimates for groundwater depth is that groundwater levels may re-equilibrate under the canal, reducing the differential between the two sides of the NMC. If that occurs, then our results should be interpreted as conservative estimates of the causal effects. To highlight this possibility, we estimate our RD model using only groundwater observations within

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the major aquifer in this region. By using observations within the same aquifer, we maximize the likelihood of groundwater depth re-equilibration in the subsample of measurements. We show these results in Column (2) of Table A3. Our results are comparable in magnitude to the main results using the full sample. Thus, even the most conservative estimates indicate a substantial difference in groundwater depth due to surface water from the NMC.

Table A3: Regression discontinuity estimates for groundwater depth with aquifer subsample

	(1)	(2)
Treatment	-23.55^{***}	-21.95^{***}
	(5.53)	(5.65)
Aquifer subsample		X
Control mean	28.74	30.28
Bandwidth (km)	9.96	9.96
Control obs.	675	631
Treatment obs.	418	360

Notes: This table reports results of estimating Equation (1) with annual average groundwater depth as the outcome variable. Groundwater depth is measured in meters below ground level (mbgl) with smaller numbers indicating groundwater that is closer to the surface. Column (1) duplicates our main result from Column (1) of Table 1. Column (2) restricts the sample of observation wells to those within the main aquifer in this region. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples and are robust to errors clustered at the observation well. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020) in Column (1) and use that same bandwidth in Column (2); we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.3.2 Additional results

Irrigation infrastructure Table 2 shows that treated villages were 11 percentage points less likely to construct an irrigation well during the years 2007–2014, saving the average village 1.37 million Indian rupees, or roughly $33,000 US dollars, in construction costs. To better understand how surface water from the NMC—which improves groundwater aquifer levels—affects the decision to invest in irrigation infrastructure, we now disaggregate these results into three distinct types of irrigation wells: deep tubewells, shallow tubewells, and dugwells.[1]

1. Tubewells access groundwater through a tube or pipe that is bored through the upper layers of the ground and into an aquifer. Deep tubewells are the deepest of these wells, reaching an underground depth of 70 meters or more,

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We first estimate RD regressions similar to those in Table 2 but using data for each specific type of irrigation well, rather than total irrigation wells. These results, which we report in Table A4, are consistent with improved groundwater availability due to the NMC, although the estimates are less precise than when we use the aggregated well data. Most strikingly, we find that the average treated village constructed 0.96 fewer deep tubewells, saving nearly 505,000 Indian rupees—more than $12,000 US dollars—both of which are substantial reductions compared to the control group mean, as shown in Columns (1) and (2). We further find some evidence of a small decrease in the number of shallow tubewells and a small increase in the number of dugwells, but these estimates are imprecise. Taken together, these results indicate that greater groundwater availability allowed farmers to avoid constructing the deepest and most costly tubewells, with some substitution to shallower and less costly wells or forgoing new well construction entirely.

Table A4: Regression discontinuity estimates for irrigation infrastructure by type

	Deep tubewell		Shallow tubewell		Dugwell	
	(1) Number	(2) Cost	(3) Number	(4) Cost	(5) Number	(6) Cost
Treatment	−0.96*	−504.9	−0.98	−41.8	0.91	−41.5
	(0.57)	(474.8)	(0.65)	(83.6)	(0.70)	(96.5)
Control mean	2.17	1689.8	1.05	151.8	3.63	449.7
Bandwidth (km)	30.37	19.48	21.31	22.48	27.87	30.29
Control obs.	1611	1066	1158	1215	1487	1607
Treatment obs.	1118	757	819	865	1038	1114

Notes: This table reports results of estimating Equation (1) with village-level measures of irrigation infrastructure construction as the outcome variable. Each pair of columns refers to a different type of well, as given in the column headers. Columns (1), (3), and (5) use the number of irrigation wells of each type constructed during the years 2007–2014. Columns (2), (4), and (6) use the total cost of constructing each type of well over these years, measured in thousands of Indian rupees. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

One feature of the disaggregated irrigation well data is that a majority of villages—even villages

and also the most costly to construct due to the depth of boring required. Shallow tubewells are less than 35 meters and, commensurately, less expensive to construct than deep tubewells. Dugwells are simply holes dug in the ground to access relatively shallow groundwater, making them the least expensive to construct.

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in the control group—did not construct a deep or shallow tubewell over these years. In other words, even in the absence of the NMC, many villages would not have constructed a deep or shallow tubewell. As a result, in a majority of locations, there is no opportunity for a village to reduce its irrigation infrastructure for these well types. To focus on locations where a response is possible, we next estimate similar RD regressions using only the non-zero observations in our dataset, effectively estimating the intensive margin effects of the NMC on irrigation infrastructure. We report these results in Table A5. These estimates are generally imprecise but provide further evidence consistent with a greater availability of groundwater. Looking at the point estimates, we find that, conditional on constructing a deep tubewell, treated villages constructed fewer deep tubewells and further reduced the costs of these wells by more than 60%. They also constructed fewer shallow tubewells and more dugwells with small changes in costs, conditional on constructing the particular well type. Thus, these intensive margin responses also align with a more easily accessible groundwater resource due to the NMC.

Table A5: Regression discontinuity estimates for irrigation infrastructure intensive margin by type

	Deep tubewell		Shallow tubewell		Dugwell	
	(1) Number	(2) Cost	(3) Number	(4) Cost	(5) Number	(6) Cost
Treatment	−2.74	−2982.9***	−1.20	−33.2	3.09	−119.1
	(1.80)	(1152.1)	(1.99)	(314.1)	(2.06)	(299.2)
Control mean	8.52	4904.7	5.79	811.8	8.35	1867.5
Bandwidth (km)	35.24	43.34	24.73	20.27	26.17	29.61
Control obs.	450	534	224	189	575	665
Treatment obs.	198	228	315	247	281	315

Notes: This table reports results of estimating Equation (1) with village-level intensive margin measures of irrigation infrastructure construction as the outcome variable. Each pair of columns refers to a different type of well, as given in the column headers. Columns (1), (3), and (5) use the number of irrigation wells of each type constructed during the years 2007–2014. Columns (2), (4), and (6) use the total cost of constructing each type of well over these years, measured in thousands of Indian rupees. In each regression, we only include a village if it constructs a non-zero number of the particular type of well during the years 2007–2014. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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Population As shown in Table 7, we find that canal surface water reduces the ambient population, or the number of people who pass through a geographic location throughout the year. We further examine if access to this resource alters the resident population, or the number of people living in a village, as reported in the decennial Census of India. To do so, we estimate our RD specification using village-level population in 2001 or 2011 as the outcome variable, and we compare the results in these years to identify any population shifts over the first decade of water regularly flowing in the NMC.

We report these results in Table A6. The estimates for total village population, in Columns (1) and (2), are imprecise, so we do not detect a statistical difference in either year. We also examine the fraction of the village from scheduled castes or tribes in order to understand if the NMC has effects on these marginalized populations. We find no effect of the NMC surface water on scheduled caste population in either year, as shown in Columns (3) and (4). Surprisingly, we do estimate that treated villages have a higher percentage from scheduled tribes, as shown in Column (5) and (6). The effect is comparable in both 2001 and 2011, however, indicating that the regular availability of surface water did not substantially alter the location of this marginalized group. Taken together, we find that the NMC did not differentially impact the population on either side of the canal.

As with the results for labor markets, nighttime lights, and ambient population in Tables 6 and 7, an important caveat is that people can easily move across the NMC. As a result, population may spill over from one side of the canal to the other. For example, workers on the treated side of the canal may reside on the control side. Then it could be that the surface water of the NMC does alter nearby population, but it equally impacts both sides of the canal, so we do not detect differential impacts in our regressions.

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Table A6: Regression discontinuity estimates for population

	Total population		Scheduled castes (%)		Scheduled tribes (%)	
	(1)	(2)	(3)	(4)	(5)	(6)
	2001	2011	2001	2011	2001	2011
Treatment	−192.0	−127.8	−0.22	−0.28	4.60**	5.24**
	(168.3)	(186.0)	(0.63)	(0.59)	(2.34)	(2.33)
Control mean	1775.6	1890.2	5.04	4.92	29.90	31.18
Bandwidth (km)	21.84	19.94	33.36	38.03	30.30	31.63
Control obs.	1484	1356	2239	2509	2034	2100
Treatment obs.	1334	1222	1886	2082	1752	1800

Notes: This table reports results of estimating Equation (1) with village-level measures of population as the outcome variable. Columns (1) and (2) use total village population, Columns (3) and (4) use percent of village in a scheduled caste, and Columns (5) and (6) use percent of village in a scheduled tribe. Columns (1), (3), and (5) use data from 2001, roughly when the NMC began flowing. Columns (2), (4), and (6) use data from a decade later in 2011. We estimate treatment effects using the robust bias-corrected RD estimator of Calonico, Cattaneo, and Titiunik (2014) with a triangular kernel; standard errors in parentheses are from the nearest-neighbor estimator for finite samples. We use the data-driven optimal bandwidth appropriate for this estimator (Calonico, Cattaneo, and Farrell 2020), and we report the number of observations within this bandwidth on each side of the NMC. The control mean reports the mean of the outcome within the bandwidth on the control side of the NMC. All regressions include covariates of soil and sedimentary deposit thickness, slope, elevation, and temperature. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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